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Electric Power Systems Research

电力系统研究

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About the journal 关于本期刊

Aims and scope 目标与范围

Electric Power Systems Research is an international medium for the publication of original papers concerned with the **generation**, **transmission**, **distribution** and **utilization** of **electrical energy**. The journal aims at presenting important results of work in this field, whether in the form of applied research, development of new procedures or components **evaluated within a system context**, original application of existing knowledge or new design approaches. The scope of *Electric Power Systems Research* is broad, yet **the submission contents should be directly related to the influence on power system operation rather than presenting the issues of particular network elements only**. **Furthermore, submissions focusing primarily on electricity prices, market economics, or isolated components without demonstrating a technical impact on the broader power system fall outside the journal's scope.**

- Systems studies including control techniques, planning, optimization methods, stability, security assessment and insulation coordination.
- Generation techniques ranging from advances in conventional electromechanical methods, through nuclear power generation, to renewable energy generation.
- Transmission, spanning the broad area from UHV (AC and DC) to network operation and protection, line routing and design.
- Distribution techniques, equipment development, and smart grids.
- Substation work: equipment design, protection and control systems.
- Utilization area from energy efficiency to distributed load levelling techniques.

Research papers and short contributions are welcome.

Review papers are accepted on invitation basis only.

Article types

Original research papers not previously published.

- Applicational and design studies.
- Comparative case studies.

Reviews

- Reviews of special topics.
- Reviews of work in progress and facilities development.
- Survey articles

Short Communications

Article length and Cover Letter:
It is compulsory to include Article Length information in the Cover Letter.
Starting with **February 20, 2026**, the Journal introduces word limits instead of page limits:

Full Length Article: max. 7000 words

Review: max. 9000 words

Short Communication: max. 1000 words

Note: All text including Abstract, tables, figures, and appendixes are counted for the word limits per guideline of each article type in below, but **references and acknowledgements are not counted**. Manuscripts exceeding word limits may be rejected before peer review.

Text Layout:

- Use single column layout, double spacing and wide (3 cm) margins on white paper. (Avoid full justification, i.e., do not use a constant right-hand margin.)
- Ensure that each new paragraph is clearly indicated.
  If possible, consult a recent issue of the journal to become familiar with its conventions.
- Number all pages consecutively; use 12 pt font size and standard fonts.

NOTE: For a LaTeX submission, you are invited to use this template.

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This journal follows a single anonymized review process. Your submission will initially be assessed by our editors. If your submission is deemed suitable, it will typically be sent to be reviewed by a minimum of two reviewers for an independent expert assessment of the scientific quality. The decision as to whether your article is accepted or rejected for publication will be made by our editors.

Read more about peer review.

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2. Drafting the article or revising it critically for important intellectual content.
3. Final approval of the version to be submitted.

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- age 年龄
- gender 性别
- race 比赛/竞赛
- ethnicity 种族/民族
- culture 文化
- sexual orientation 性取向
- disability or health condition

  残疾或健康状况

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我们建议避免使用与个人特征相关的描述性词汇，除非这些描述确实必要且恰当。在写作时，应默认使用复数形式来体现中立性（例如“临床医生、患者/客户”）。只要有可能，应避免使用“他/她”这样的单数形式来指代人物。

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These guidelines are meant as a point of reference to help you identify appropriate language but are by no means exhaustive or definitive.

这些准则旨在作为参考，帮助您选择恰当的用词。不过，这些准则绝非面面俱到或具有绝对的权威性。

Reporting sex- and gender-based analyses

报告基于性别和性别的分析结果

There is no single, universally agreed-upon set of guidelines for defining sex and gender. We offer the following guidance:

对于如何定义“性别”这一概念，目前并没有一套被所有人所认同的统一标准。我们提出以下建议作为参考：

- Sex and gender-based analyses (SGBA) should be integrated into research design when research involves or pertains to humans, animals or eukaryotic cells. This should be done in accordance with any requirements set by funders or sponsors and best practices within a field.

  当研究涉及人类、动物或真核细胞时，应将基于性别和性别的分析方法纳入研究设计中。这一做法应符合资金提供者或赞助方的要求，同时也应遵循该领域的最佳实践。

- Sex and/or gender dimensions of the research should be addressed within the article or declared as a limitation to the generalizability of the research.

  研究中涉及性别方面的内容，应在文章中予以说明，或者将此视为该研究结果普遍适用性的限制因素加以说明。

- Definitions of sex and/or gender applied should be explicitly stated to enhance the precision, rigor and reproducibility of the research and to avoid ambiguity or conflation of terms and the constructs to which they refer.

  在研究中，必须明确界定“性别”和/或“性别的含义”，这样才能提高研究的精确性、严谨性及可重复性。同时，这样做也有助于避免术语及其所代表的概念之间的混淆或歧义。

We advise you to read the Sex and Gender Equity in Research (SAGER) guidelines and the SAGER checklist (PDF) on the EASE website, which offer systematic approaches to the use of sex and gender information in study design, data analysis, outcome reporting and research interpretation.

我们建议您阅读 EASE 网站上的《研究中的性别平等准则》以及相关的检查清单（PDF 格式）。这些资料为如何在研究设计、数据分析、结果报告和研究解读中妥善运用性别相关数据提供了系统的指导。

For further information we suggest reading the rationale behind and recommended use of the SAGER guidelines.

如需了解更多信息，建议您仔细阅读有关 SAGER 指南的制定依据及推荐使用方式。

Definitions of sex and/or gender

性别与/或性别的定义

We ask authors to define how sex and gender have been used in their research and publication. Some guidance:

我们要求作者说明在他们的研究和著作中，是如何使用“性别”和“性别认同”这些概念的。以下是一些指导原则：

- Sex generally refers to a set of biological attributes that are associated with physical and physiological features such as chromosomal genotype, hormonal levels, internal and external anatomy. A binary sex categorization (male/female) is usually designated at birth ("sex assigned at birth") and is in most cases based solely on the visible external anatomy of a newborn. In reality, sex categorizations include people who are intersex/have differences of sex development (DSD).

  所谓“性别”，通常指的是与染色体基因型、激素水平以及身体内外部解剖结构等生理特征相关的一系列生物学属性。二元的性别分类（男性/女性）通常在出生时就被确定下来，而这一分类主要基于新生儿的外部解剖特征。实际上，性别分类还应包括那些属于双性人或存在性别发育异常的人。

- Gender generally refers to socially constructed roles, behaviors and identities of women, men and gender-diverse people that occur in a historical and cultural context and may vary across societies and over time. Gender influences how people view themselves and each other, how they behave and interact and how power is distributed in society.

  性别通常指的是在历史和文化背景下，社会对女性、男性和各种性别群体所规定的角色、行为和身份认同。这些规范因社会和时代的不同而有所差异。性别影响着人们如何看待自己和他人、如何行事和互动，也影响着社会中的权力分配方式。

Depending on the focus of a paper, sex and/or gender may or may not be relevant to the content of the paper. We recognize that beliefs, attitudes, and laws relating to sex and gender may vary. These articles do not attempt to dictate author beliefs but rather require that, where relevant to an author's research or paper, the author must provide clear explanations of how the paper and research define and use sex and gender.

根据论文的主题不同，性别因素可能与之相关，也可能无关。我们认识到，关于性别和性别的各种观念、态度和法律规范可能因地区而异。这些准则并不试图强制作者必须持有某种特定的观点，而是要求：如果性别因素与作者的研究或论文相关，那么作者必须明确说明在论文中是如何定义和运用性别这一概念的。

Image manipulation 图像处理/图像编辑

We accept that authors sometimes need to adjust images for clarity but any manipulation of images for the purpose of deception or fraud will be seen as scientific ethical abuse and will be dealt with accordingly.

我们承认，作者有时需要调整图片以提升其清晰度。但是，任何出于欺骗或欺诈目的而对图片进行的篡改行为，都将被视为违反科学伦理的行为，我们将对此予以相应的处理。

Authors must adhere to this journal's policy for graphical images:

作者必须遵守本期刊关于图形图像的使用规定：

- No specific feature within an image may be enhanced, obscured, moved, removed or introduced.

  图像中的任何特定元素都不可以被增强、隐藏、移动、删除或添加。

- Adjustments of brightness, contrast, or color balance are acceptable if, and only as long as, they do not obscure or eliminate any information present in the original image.

  只要对亮度、对比度或色彩平衡的调整不会使原始图像中的任何信息变得难以辨认或完全消失，那么这样的调整就是可以接受的。

- Nonlinear adjustments such as changes to gamma settings must be disclosed in the figure legend.

  诸如伽马值调整之类的非线性调整措施，必须在该图的图例中予以说明。

- Use of AI tools to create or manipulate primary research images is not permitted. This includes images that represent experimental or observational data, such as western blots, microscopy images, histology images, or patient images. Please read our policy on the use of

AI tools in figures, images and artwork, which can be found in Elsevier's GenAI Policies for Journals.

禁止使用人工智能工具来创建或修改用于原始研究的图像。这包括所有能够体现实验或观测数据的图像，比如 Western 印迹图、显微镜图像、组织学图像以及患者影像等。请仔细阅读我们在《Elsevier 期刊的 GenAI 政策》中关于在图表、图像和艺术作品中使用人工智能工具的详细规定。

To verify compliance with the above, this journal may send your images to a secure third-party tool to enable screening for image irregularities. Our editors may ask you to provide original data or images if any questions arise as a result of the screening. The final decision as to whether images are acceptable will be taken by our editors.

为了确保符合上述要求，本期刊可能会将您的图片发送给第三方安全机构，以检测图片是否存在异常。如果检测过程中出现任何问题，编辑们可能会要求您提供原始数据或图片。最终是否接受这些图片，由编辑们来决定。

Authors are encouraged to carefully check all images before submission and to connect all the data in any figures to the original, unprocessed data.

建议作者在提交稿件之前仔细检查所有图片，并确保图表中的所有数据都与原始的、未经处理的数据相对应。

Jurisdictional claims 管辖权方面的主张/关于管辖权的争议

Elsevier respects the decisions taken by its authors as to how they choose to designate territories and identify their affiliations in their published content. Elsevier's policy is to take a neutral position with respect to territorial disputes or jurisdictional claims, including, but not limited to, maps and institutional affiliations. For journals that Elsevier publishes on behalf of a third party owner, the owner may set its own policy on these issues.

爱思唯尔尊重作者们在确定所发表内容的地域归属及机构隶属关系时的决定。在处理与地域划分或管辖权相关的问题时，爱思唯尔采取中立立场，这包括地图的标注以及机构信息的呈现方式等。对于那些由爱思唯尔代表第三方所有者来出版的期刊，其所有者可自行制定相关规则。

- Maps: Readers should be able to locate any study areas shown within maps using common mapping platforms. Maps should only show the area actually studied and authors should not include a location map which displays a larger area than the bounding box of the study area. Authors should add a note clearly stating that "*map lines delineate study areas and do not necessarily depict accepted national boundaries*". During the review process, Elsevier's editors may request authors to change maps if these guidelines are not followed.

  地图：读者应能够使用常见的地图查看工具，找到地图上标出的所有研究区域。地图上应仅显示实际被研究的区域；作者不应绘制出超出研究区域边界的地图。作者需在地图上注明："地图上的边界线仅用于标明研究区域，并不一定代表公认的国家边界"。在审稿过程中，如果作者没有遵守这些准则，Elsevier 的编辑可能会要求其修改地图。

- Institutional affiliations: Authors should use either the full, standard title of their institution or the standard abbreviation of the institutional name so that the institutional name can be independently verified for research integrity purposes.

  机构隶属关系：作者应使用所在机构的完整正式名称或标准缩写，以便能够独立核实该机构的真实性，从而确保研究的可靠性。

Delayed publication 延迟发布

If you need to delay the publication of your article for any reason, please contact the editorial office of the journal or our Journal Article Publishing Support Center at the earliest possible opportunity. While we will try to ensure that your article is not published before a certain date if you make your request on a timely basis, we cannot guarantee this. Please also note that once the agreed period of delay has expired the article may be published at any time thereafter.

如果您因任何原因需要推迟文章的发表，请尽快联系该期刊的编辑部或我们的文章发表支持中心。如果您能及时提出请求，我们会尽力确保您的文章不会在某个特定日期之前被发表。不过，我们无法完全保证这一点。另外，请注意，一旦约定的延迟期限过后，文章随时都可能被发表。

Withdrawal, correction, or retraction

撤回、更正或撤销

For questions about article withdrawal, correction, or retraction, please consult Elsevier's Article Correction, Retraction and Removal Policy.

如有关于文章撤回、更正或撤销的疑问，请参阅爱思唯尔的《文章更正、撤回与删除政策》。

Writing and formatting 写作与格式编排

File format 文件格式

We ask you to provide editable source files for your entire submission (including figures, tables and text graphics). Some guidelines:

我们要求您提供所有提交内容的可编辑源文件，包括图表、表格和文字内容。具体要求如下：

- Save files in an editable format, using the extension .doc/.docx for Word files and .tex for LaTeX files. A PDF is not an acceptable source file.

  请将文件保存为可编辑格式：Word 文档请使用.doc/.docx 格式，LaTeX 文档请使用.tex 格式。PDF 格式的文件不可作为输入文件使用。

- Format Word files in a single-column layout. Double-column formatting is only permitted for LaTeX submissions.

  将 Word 文档设置为单列布局。双列布局仅适用于以 LaTeX 格式提交的文档。

- Remove any strikethrough and underlined text from your manuscript, unless it has scientific significance related to your article.

  请删除手稿中所有被划线或加下划线的文字，除非这些文字与您的文章具有科学上的重要性。

- Use spell-check and grammar-check functions to avoid errors.

  请使用拼写检查和语法检查功能，以避免出现错误。

We advise you to read our Step-by-step guide to publishing with Elsevier.

我们建议您阅读我们的《在 Elsevier 上发表论文的详细指南》。

LaTeX

We encourage you use our LaTeX template when preparing a LaTeX submission. You will be asked to provide all relevant editable source files upon submission or revision.

我们建议您在用 LaTeX 格式提交文档时使用我们的模板。在提交或修改文档时，您需要提供所有可编辑的源文件。

Support for your LaTeX submission:

对您提交的 LaTeX 文档的支持：

- LaTeX submission instructions and templates

  LaTeX 投稿指南与模板

- Journal Article Publishing Support Center LaTeX FAQs and support

  期刊文章发表支持中心：LaTeX 相关常见问题与支持服务

- Researcher Academy's Beginners' guide to writing a manuscript in LaTeX

  研究员学院：用 LaTeX 撰写手稿的入门指南

Title page 标题页

You are required to include the following details on the title page:

您必须在标题页上注明以下详细信息：

- Article title. Article titles should be concise and informative. Please avoid abbreviations and formulae, where possible, unless they are established and widely understood, e.g. DNA.

  文章标题。文章标题应简洁明了且能准确传达文章内容。请尽量避免使用缩写和公式，除非这些缩写或公式已被广泛接受和理解，比如 DNA。

- Author names. Provide the given name(s) and family name(s) of each author. The order of authors should match the order in the submission system. Carefully check that all names are accurately spelled. You may add your name between parentheses in your own

script after the English transliteration.

作者姓名。请列出每位作者的姓名和姓氏。作者的顺序应与提交系统中的顺序一致。请仔细核对所有姓名的拼写是否正确。您可以在英文姓名的转写之后，用自己的文字把姓名加上，用括号括起来。

- Affiliations, referring to where the work was carried out, below the author names. Indicate affiliations using a lower-case superscript letter immediately after the author's name and in front of the corresponding address. Ensure that you provide the full name and postal address of each affiliation, including the country name.

  作者姓名下方应注明研究工作的开展地点。具体信息需以小写字母的上标形式标注在作者姓名之后、相应地址之前。请务必提供每个研究机构的完整名称和邮政地址，包括所在国家/地区的名称。

- Corresponding author. Clearly indicate who will handle correspondence for your article at all stages of the refereeing and publication process and also post-publication. This responsibility includes answering any future queries about your results, data, or methodology. The email address of the corresponding author must be included on the title page and will be published in the article.

  通讯作者。请明确指定在论文的整个审稿和发表过程中，以及发表后的所有环节中，谁将负责处理与本文相关的所有通信事宜。这一职责还包括回答有关论文结果、数据或研究方法的任何疑问。通讯作者的电子邮件地址必须列在论文的标题页上，并会同时发表在论文中。

- Present/permanent address. If an author has changed affiliation since the work described in your article was carried out, or was visiting during that time, a "present address" (or "permanent address") can be indicated by a footnote to the author's name. The address where the author carried out the work must be retained as their main affiliation address. Use superscript Arabic numerals for such footnotes.

  现址/永久地址。如果作者在撰写文章时所从事的研究工作是在另一机构完成的，或者当时只是作为访问学者在该机构工作，那么可以在作者姓名的脚注中注明“现址”或“永久地址”。不过，作者实际从事研究工作的机构地址仍应作为其主要联系方式。此类脚注应使用上标阿拉伯数字来标注。

Abstract 摘要

You are required to provide a concise and factual abstract which does not exceed 250 words. The abstract should:

您需要提供一份简明扼要、内容真实的摘要，字数不得超过 250 字。摘要应具备以下特点：

- State the purpose of your paper.

  请说明你撰写这篇论文的目的。

- Outline basic procedures followed.

  概述所遵循的基本程序。

- Include your main findings, their statistical significance, where applicable, and your principal conclusions.

  请列出你的主要研究结果、这些结果的统计显著性（如适用），以及你的最终结论。

- Emphasize new and important aspects of your study or observations.

  请着重强调你的研究或观察结果中那些新颖且重要的方面。

Some guidelines: 一些指导原则：

- Abstracts must be able to stand alone, as abstracts are often presented separately from the article.

  摘要应当具有独立的完整性，因为摘要通常是与文章分开来呈现的。

- Avoid references. If any are essential to include, ensure that you cite the author(s) and year(s).

  避免使用引用内容。如果确实有必要引用，请务必注明作者和引用年份。

- Avoid non-standard or uncommon abbreviations. If any are essential to include, ensure they are defined within your abstract at first mention.

Keywords

You are required to provide 1 to 7 keywords for indexing purposes. Keywords should be written in English. Please try to avoid keywords consisting of multiple words (using "and" or "of").

We recommend that you only use abbreviations in keywords if they are firmly established in the field.

Highlights

You are required to provide article highlights at submission.

Highlights are a short collection of bullet points that should capture the novel results of your research as well as any new methods used during your study. Highlights will help increase the discoverability of your article via search engines. Some guidelines:

- Submit highlights as a separate editable file in the online submission system with the word "highlights" included in the file name.
- Highlights should consist of 3 to 5 bullet points, each a maximum of 85 characters, including spaces.

We encourage you to view example article highlights and read about the benefits of their inclusion.

Graphical abstract

You are encouraged to provide a graphical abstract at submission.

The graphical abstract summarizes the contents of your article in a concise, pictorial, professional form, highlighting important aspects of the research. It is designed to appeal to an interdisciplinary audience, and draw more attention to your online article. Some guidelines:

- Ensure you have obtained the necessary permission to include any third party owned material in a graphical abstract.
- Use of generative AI or AI-assisted tools in the production of graphical abstracts must align with Elsevier's GenAI Policies for Journals.
- Submit your graphical abstract as a separate file in the online submission system.
- Ensure the image is 531 x 1328 pixels (h x w) or proportionally more, and is readable at a size of 5 x 13 cm.
- Preferred file types for graphical abstracts are TIFF, EPS, PDF or MS Office files.

More information, examples and a template can be found in the graphical abstract guidelines.

Math formulae

- Submit math equations as editable text, not as images.
- Present simple formulae in line with normal text, where possible.
- Use the solidus (/) instead of a horizontal line for small fractional terms such as X/Y.
- Present variables in italics.
- Denote powers of e by exp.
- Display equations separately from your text, numbering them consecutively in the order they are referred to within your text.

Tables

Tables must be submitted as editable text, not as images. Some guidelines:

- Place tables next to the relevant text or on a separate page(s) at the end of your article.
- Cite all tables in the manuscript text.
- Number tables consecutively according to their appearance in the text.
- Please provide captions along with the tables.

- Place any table notes below the table body.
- Avoid vertical rules and shading within table cells.

We recommend that you use tables sparingly, ensuring that any data presented in tables is not duplicating results described elsewhere in the article.

Figures, images and other artwork

Figures, images, diagrams and other graphical media (artwork) must be supplied as separate files along with the manuscript, in line with these artwork and media instructions. Some excerpts:

When submitting artwork:

- Cite all images in the manuscript text.
- Number images according to the order they appear within your article.
- Submit each image as a separate file using a logical naming convention for your files (for example, Figure_1, Figure_2 etc).
- Text graphics may be embedded in the text at the appropriate position. If you are working with LaTeX, text graphics may also be embedded in the file.

Captions

All artwork must have a caption. A caption should consist of a brief title (not displayed on the figure itself) and a description of the image. Keep the amount of text to a minimum, though any symbol or abbreviation should be explained.

Read how to add captions to your submission here.

Formats

Submit your artwork in the formats listed below, following the given requirements:

Image type	Recommended file type	Requirements
Vector drawings	EPS, PDF	Embed the font or save text as graphics
Photographs (color or grayscale)	TIFF, JPG, PNG	Minimum resolution 300 dpi (min. width: single column: 1063 pixels; full page: 2244 pixels)
Bitmapped line drawings	TIFF, JPG, PNG	Minimum resolution 1000 dpi (min. width: single column: 3543 pixels; full page: 7480 pixels)
Charts and large or shaded tables	XLS, XLSX	Supply in a clearly readable format
Annotated images	PPT, PPTX	Annotations and text must be legible; Minimum resolution of embedded images 300 dpi

Do not submit:

- Files that are too low in resolution.

- Artwork where text is disproportionately small compared to the image size, as text may become unreadable.
- Multiple images or graphs combined into one file, as this affects legibility and accessibility. An exception may be made for images that are intended to be compared to each other, such as before, during, and after photographs.

Color artwork

If you submit usable color figures with your accepted article, we will ensure that they appear in color online. Please ensure that color images are accessible to all, including those with impaired color vision. Learn more about color and web accessibility.

Generative AI and figures, images and other artwork

Please read our policy on the use of generative AI and AI-assisted tools in figures, images and artwork, which can be found in Elsevier's GenAI Policies for Journals. This policy states:

- AI tools may be used to support the creation of explanatory images (such as flow charts, conceptual diagrams, or schematic illustrations)
- AI tools may be used to create data visualizations (such as plots, charts, graphs, or heatmaps) when the output is directly derived from underlying data using reproducible analytical, computational, or statistical methods.
- AI tools must not be used to create or alter images that represent primary observed or experimental data (such as western blots, microscopy images, histology images, or patient images) that were not directly obtained in the research. This includes adjustments to brightness, contrast, or color balance, which should only be done using established image processing software.
- This policy does not prevent the use of AI tools in formal research design or research methods (such as in AI-assisted imaging approaches used for collection and/or interpretation of the underlying research data or predictive modelling of clinical outcomes based on patient data).

Authors are responsible for ensuring the accuracy and originality of all images submitted for publication. If you used generative AI tools to create images in your manuscript, you should disclose this in each image caption as well as in the general Generative AI disclosure statement.

Supplementary material

We encourage the use of supplementary materials such as applications, images and sound clips to enhance research. Some guidelines:

- Supplementary material should be accurate and relevant to the research.
- Cite all supplementary files in the manuscript text.
- Submit all supplementary materials at the same time as your manuscript.
- Include a concise, descriptive caption for each supplementary file, describing its content.
- After submission supplementary files can only be added or replaced in the revision stage of the editorial process.
- Be aware that all supplementary materials provided will appear online in the exact same way as received. These files will not be checked, formatted or typeset by the production team.

Video

This journal accepts video material and animation sequences to support and enhance your scientific research. We encourage you to include links to video or animation files within articles. Some guidelines:

- When including video or animation file links within your article, refer to the video or animation content by adding a note in your text where the file should be placed.
- Clearly label files ensuring the given file name is directly related to the file content.
- Provide files in one of our recommended file formats. Files should be within our preferred maximum file size of 150 MB per file, 1 GB in total.

- Provide "stills" for each of your files. These will be used as standard icons to personalize the link to your video data. You can choose any frame from your video or animation or make a separate image.

- Provide descriptive text in your manuscript to refer to the video content. This text helps ensure accessibility for visually impaired readers who rely on descriptive information. For journals publishing in print this is also essential, as video and animation files cannot be embedded in the print version.

We publish all video and animation files supplied in the electronic version of your article.

For more detailed instructions, we recommend that you read our guidelines on submitting video content to be included in the body of an article.

Research data

We are committed to supporting the storage of, access to and discovery of research data, and our research data policy sets out the principles guiding how we work with the research community to support a more efficient and transparent research process.

Research data refers to the results of observations or experimentation that validate research findings, which may also include software, code, models, algorithms, protocols, methods and other useful materials related to the project.

Please read our guidelines on sharing research data for more information on depositing, sharing and using research data and other relevant research materials.

Research data deposit, citation and linking

For this journal, **Option C** instructions from our research data guidelines apply. This means that you are **required** to:

- Deposit your research data in a relevant data repository.

- Cite and link to this dataset in your article.

- If this is not possible, make a statement explaining why research data cannot be shared.

Data statement

To foster transparency, you are required to state the availability of any data at submission.

Ensuring data is available may be a requirement of your funding body or institution. If your data is unavailable to access or unsuitable to post, you can state the reason why (e.g., your research data includes sensitive or confidential information such as patient data) during the submission process. This statement will appear with your published article on ScienceDirect.

Read more about the importance and benefits of providing a data statement.

Data linking

Linking to the data underlying your work increases your exposure and may lead to new collaborations. It also provides readers with a better understanding of the described research.

If your research data has been made available in a data repository there are a number of ways your article can be linked directly to the dataset:

- Provide a link to your dataset when prompted during the online submission process.

- For some data repositories, a repository banner will automatically appear next to your published article on ScienceDirect.

- You can also link relevant data or entities within the text of your article through the use of identifiers. Use the following format: Database: 12345 (e.g. TAIR: AT1G01020; CCDC: 734053; PDB: 1XFN).

Learn more about linking research data and research articles in ScienceDirect.

Co-submission of related data, methods or protocols

You are encouraged to publish a description of your research data, methods, or protocols related to your regular article as a co-submission article in Data in Brief or MethodsX.

This co-submission can be submitted at the same time as your regular manuscript submission to this journal. If both your regular and co-submitted manuscripts are accepted for publication, they will be linked together on ScienceDirect, thereby promoting research reproducibility, interoperability, and open science.

- Prepare a separate manuscript describing your research data, methods, or protocols using the mandatory templates from Data in Brief or MethodsX.
- Submit this co-submission alongside your regular manuscript as a separate submission item; on the 'Attach files' page in Editorial Manager, after which it will be automatically forwarded to the *Data in Brief* or *MethodsX* and assigned an editor.
- For associated Article Processing Charge (APC) details, please visit:
  - MethodsX
  - Data in Brief

Please note: Due to the automated process of submitting to *Data in Brief* or *MethodsX*, only after acceptance will you be presented with a personalized OA Article Publishing Charge based on your individual context (your country, institutional affiliation, and any society membership for example).

Please visit the co-submission Researcher Support page for more information on how to submit a co-submission.

Article structure

Article sections

- Divide your manuscript into clearly defined and numbered sections. Number subsections 1.1 (then 1.1.1, 1.1.2, ...), then 1.2, etc.
- Use the numbering format when cross-referencing within your article. Do not just refer to "the text."
- You may give subsections a brief heading. Headings should appear on a separate line.
- Do not include the article abstract within section numbering.

Theory and calculation

The theory section should lay the foundation for further work by extending the background you provided in the introduction to your article. The calculation section should represent a practical development from a theoretical basis.

Acknowledgements

Include any individuals who provided you with help during your research, such as help with language, writing or proof reading, in the acknowledgements section.

Acknowledgements should be placed in a separate section which appears directly before the reference list. Do not include acknowledgements on your title page, as a footnote to your title, or anywhere else in your article other than in the separate acknowledgements section.

Author contributions: CRediT

Corresponding authors are required to acknowledge co-author contributions using CRediT (Contributor Roles Taxonomy) roles:

- Conceptualization
- Data curation
- Formal analysis
- Funding acquisition
- Investigation
- Methodology

- Project administration
- Resources
- Software
- Supervision
- Validation
- Visualization
- Writing – original draft
- Writing – review and editing

Not all CRediT roles will apply to every manuscript and some authors may contribute through multiple roles.

We advise you to read more about CRediT and view an example of a CRediT author statement.

Appendices
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[3] W. Strunk Jr., E.B. White, The Elements of Style, fourth ed., Longman, New York, 2000.

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Reference to a website:
[5] Cancer Research UK, Cancer statistics reports for the UK. http://www.cancerresearchuk.org/aboutcancer/statistics/cancerstatsreport/, 2023 (accessed 13 March 2023).

Reference to a dataset:
[6] M. Oguro, S. Imahiro, S. Saito, T. Nakashizuka, Mortality data for Japanese oak wilt disease and surrounding forest compositions [dataset], Mendeley Data, v1, 2015. https://doi.org/10.1234/abc12nb39r.1.

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[7] E. Coon, M. Berndt, A. Jan, D. Svyatsky, A. Atchley, E. Kikinzon, D. Harp, G. Manzini, E. Shelef, K. Lipnikov, R. Garimella, C. Xu, D. Moulton, S. Karra, S. Painter, E. Jafarov, S. Molins, Advanced Terrestrial Simulator (ATS) v0.88 [software], Zenodo, March 25, 2020. https://doi.org/10.1234/zenodo.3727209.

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